

Mindmap

Python Crash Course

Status: September 23, 2025

Operators

Mathematical

+ - * / ** // %

Assignment

= += -= *= /= **= %=

Logical

> < >= <= == != and or not

Bitwise logical

& |

Functions

Built-in

type()
isinstance()
print()
id()
round()
input()
open()
len()
all()
any()
sorted()

Built-in methods

.copy()
.apply()

Built-in set/list/tuple methods

.add()
.remove()
.pop()
.count()
.index()
.sort()
.reverse()

Built-in dict methods

.keys()
.values()
.items()

CLI commands

	windows	macOS/linux
Print out current working directory	chdir	pwd
Print out files in current working directory	dir	ls
Change working directory	cd <folderpath>	
Run python script	python <filepath>	
Open Python interpreter in CLI	python	
Exit Python interpreter in CLI	exit() + ENTER	

Built-in modules

math
random
os
re
time
collections
operator
csv
json

Python packages

pandas
matplotlib.pyplot

Built-in str methods

.upper()
.lower()
.split()

Built-in i/o methods

.read()
.readlines()
.write()
.writelines()

Keywords

if elif else and
not or True
False def return
in assert for
while break del
raise pass
import as from
None with try
except lambda

Coding environment & coding concepts

jupyter notebook interface

Markdown language

#comments in code

assigning a value to a variable

concatenating strings

calling a function

indexing & slicing (for str, list)

control flow

conditional statements

for loops while loops

assert statements

try/except (catching errors)

defining & calling your own functions

variable scope: global & local

arguments & parameters (required vs. optional)

string formatting f'{}' r'' \n \t

regular expressions

GUI (graphical user interface) vs. CLI (command line interface)

absolute vs. relative file paths

“Magic” (% for inline and %% for cell magic) in Jupyter notebook

%%run <filepath> for running a Python file in Jupyter notebook

inline functions with lambda

list comprehension

type checking

random seeding

%%time and %%timeit for timing Jupyter notebook cell runtime

01/01/1970: Unix time starts

Conda for environment management

Monte Carlo simulations (random sampling)

Writing your own modules (.py files with function definitions)

pandas functions

```
pd.read_csv()  
pd.DataFrame()  
pd.concat()
```

pandas attributes (for DataFrames)

```
.index  
.column  
.shape
```

pandas methods (for DataFrames)

```
.head()  
.tail()  
.describe()  
.sort_values()  
.drop()  
.reset_index()  
.unique()  
.isna()  
.notna()  
.fillna()  
.replace()  
.groupby()  
.copy()
```

pandas operators

```
~ (tilde - inversion)
```


OPERATORS

+
-
*
/
**
=

FUNCTIONS

type()

print()

1/2
3/4
x=5

DATA TYPES

int
float
str

SOFTWARE

error
message
markdown
#comments
assigning
concatenate

Aug 26, 2025

Aug 28, 2025

OPERATORS

math $+$ $-$ $/$ $*$ $**$ $//$ $\%$
assign $=$
logical $<$ $>$ $==$ $!=$ and
 $>=$ $<=$ not or

DATA TYPES

int

float

str

bool

List

True

False

METHODS. ()
FUNCTIONS
round() type() len()
sorted() print()
id() { int() bool()
all() { str() list()
any() { float()

Sep 2, 2025

OPERATORS

assignment =

$\neq$

$+=$

$\% =$

$- =$

$/ =$

?

D.T.

tuple

set

()
{ }

SCOPE

KEYWORDS

and

True

or

False

not

if

in

else

def

elif

return

B.i.F

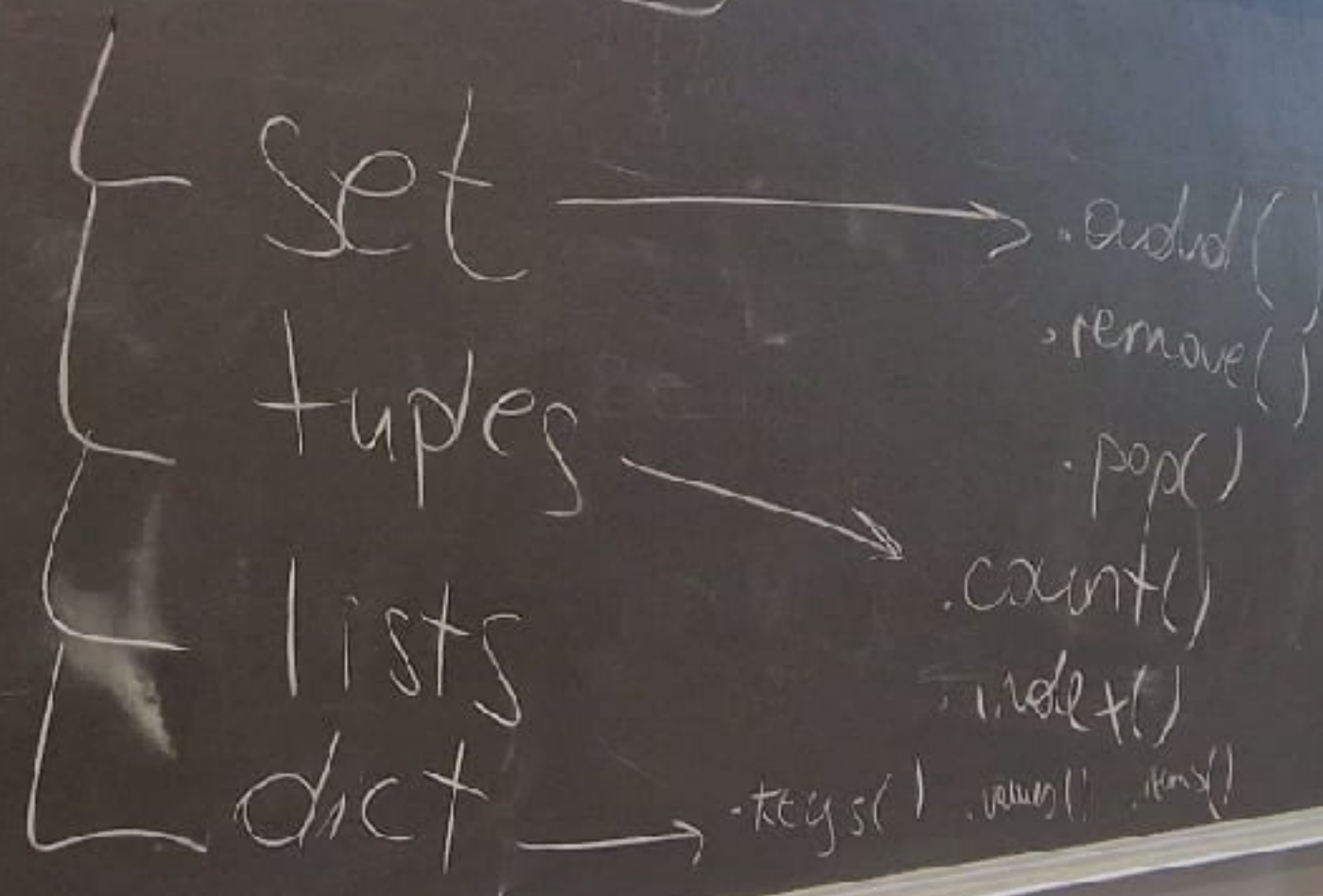
tuple()

list()

set() $\rightarrow$.add()

Sep 4, 2025

METHODS



While cond:
statement

B.F
dict()
range()

FLOW
for

in O::

assert
DT set
range
dict { }
K for
while
break
del
PAIRS
in

Sep 9, 2025

DATA
TYPES
None

KEY/WORDS

raise
assert
pass

as

None
import
from

METHODS

for str:

.upper()

.lower()

.split()

etc.

MODULES

math

random

os

PACKAGES

pandas

plt

LINGO

FUNCTION

parameter

argument

definition

call

f"{}"

sqrt()

choice()

listdir()

Sep 11, 2025

TEXT
FORMAT
f"{}"
\\n r" "
\\t r" "
RAW
l=
== >

BIF:
open()
input()
path / abs
rel

CODING STUFF
shell
CLI vs GUI
%orun
%%

WIN	MAC
chdir	pwd
dir	ls
cd	cd

↑ ..

module
re
• search
• findall

(KEY)
with

Sep 16, 2025

CONTROL
FLOW

try
except

KEYWORDS
try except
lambda

OPERATORS

& |

BUILT-IN MODULES

CSV JSON

PANDAS

.loc[]

.concat()

CODING STYLES

list comprehension

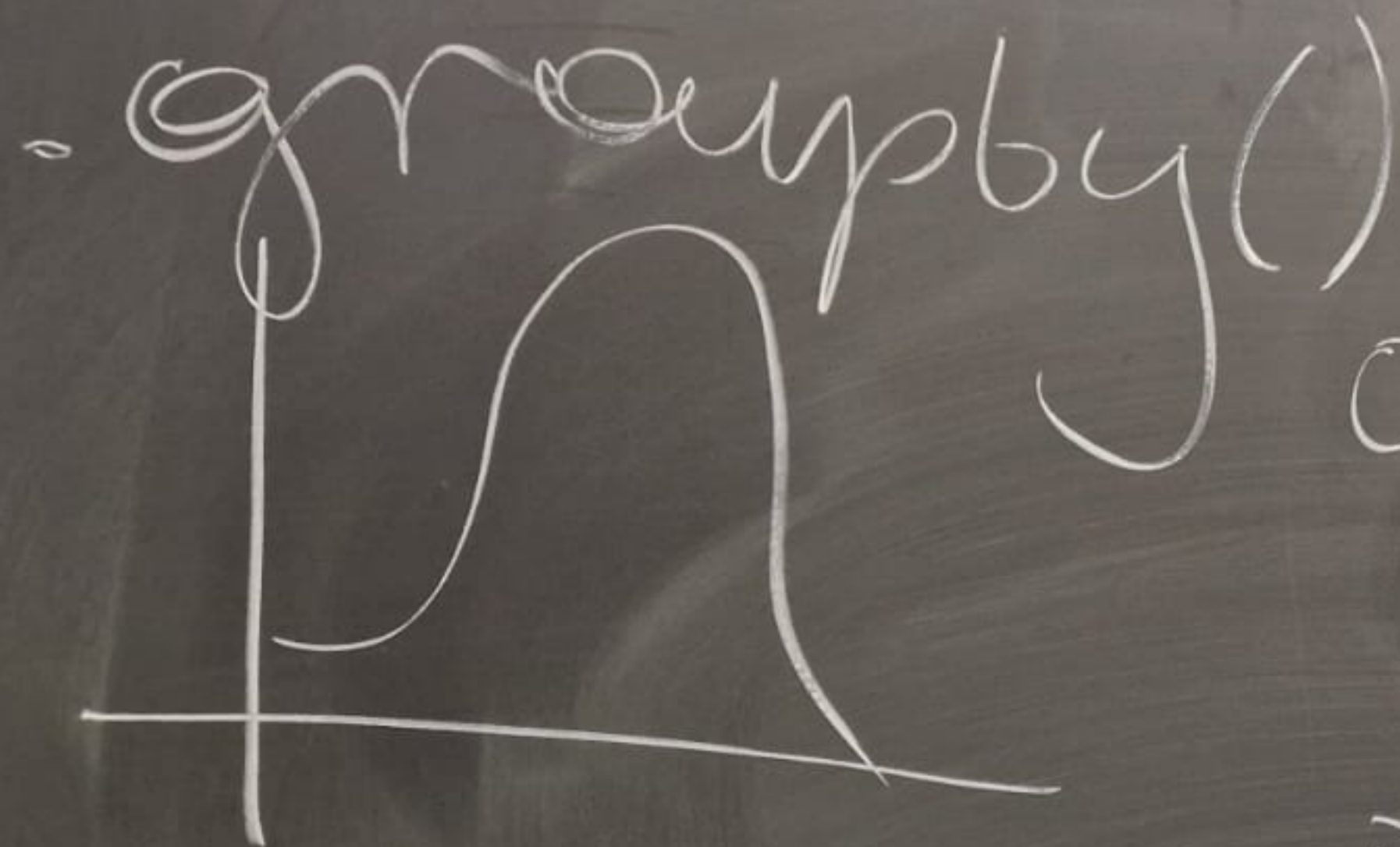
attributes

.apply()

lambda

Sep 18, 2025

MORE
PANDAS



BIF

isinstance()

CODING
HACKS

OPERATOR

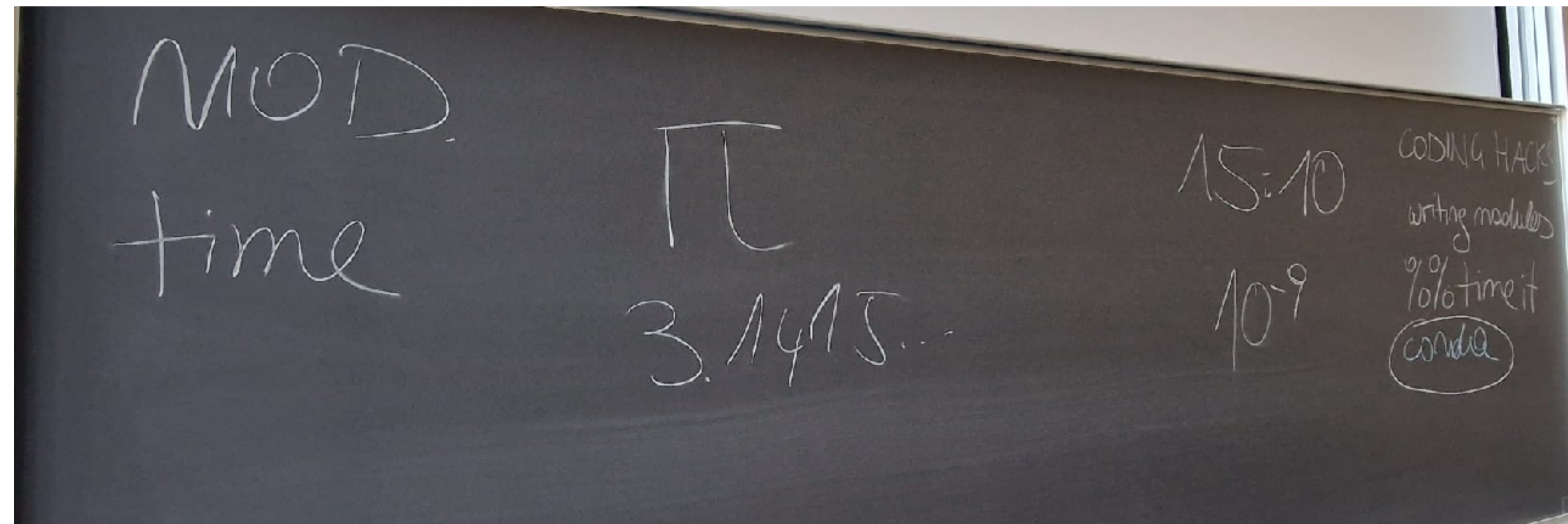
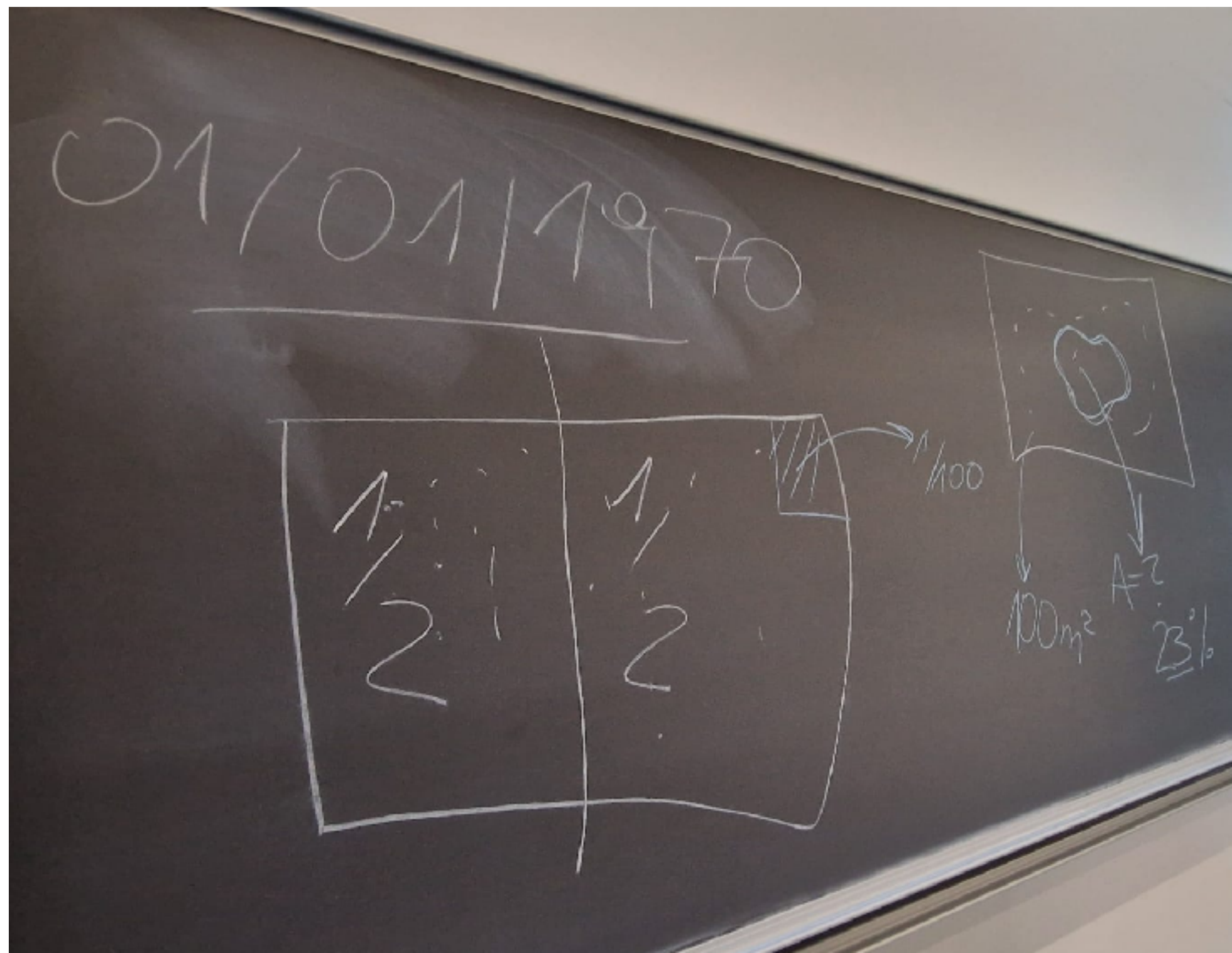
N

BINARY
SEARCH

DTYPE

(NAN) NA

Sep 23, 2025



Sep 23, 2025

$$\frac{A_c}{A_{SQ}} = \frac{\pi}{4} \longrightarrow \pi = \frac{4 \cdot A_c}{A_{SQ}}$$

$$\frac{A_{\square}}{A_{\square SQ}} = \frac{\frac{\pi}{4}}{1} = \frac{\pi}{4}$$

$$1 \text{ m}^2$$

$$\frac{r^2 \pi}{4} = \frac{\pi}{4}$$

